

In this equation, we suppress any notation to indicate that this regression applies at any and all times. For example, without loss of generality, we could subscript all the variables and parameters in the equation with a time index.

The ε_i terms, for all $i = 1, \dots, N$ are the terms that represent the richness and cheapness of the bonds in our data set, with positive values indicating the bond in question as rich.

Because $A(\Theta_i; \alpha)$ is a function involving the cash flows of the coupon bond, and because the value of a coupon bond is a linear function of the values of the individual cash flows, $A(\Theta_i; \alpha)$ generally will have a form given by:

$$A(\Theta_i; \alpha) = \sum_{l=1}^{M_i} C_{il} \phi(\tau_{il})$$

where C_{il} is the size of the i -th cash flow for bond i , and $\phi(\tau_{il})$ is the discount factor associated with the i -th cash flow for the i -th bond, to be received at time τ_{il} .

With this in mind, we see that the key to taking a view as to the richness or cheapness of specific bonds among a collection of bonds is to ascertain an appropriate discount function $\phi(\tau)$ that is presumed to apply generally to the collection of bonds at any particular point in time.

As with standard regressions, we can proceed once we've made some distributional assumptions on the error terms. In particular, we'll assume that the error terms in the equation for P_i are independent of one another and that they're normally distributed, each with a mean equal to zero. For now, we'll also assume that all the error terms have variances that are identical to each other. In this case, the maximum likelihood estimators for α and γ are identical to the estimators obtained via ordinary least squares (OLS). If we have reason to believe that the error terms are in some sense heteroscedastic, we can use weighted least squares (WLS) in place of OLS.

To make the regression operational, we need to specify two more aspects of our regression. First, we need to decide on a functional form for the discount factor function $\phi(\tau)$. Second, we need to specify the independent variables on which we believe the bond prices are conditioned.

SPECIFYING A FUNCTION FOR DISCOUNT FACTORS

We use the term *discount factor* to refer to the price of a zero-coupon bond as a percentage of its face value. In the context of our exercise, we assume that at any point in time there exists one and only one discount factor corresponding to every future date on which a zero-coupon bond could mature. An absence of

arbitrage ensures the uniqueness of each discount factor and that all discount factors are non-negative.

In the past, it was common to make the additional assumption that all discount factors were no greater than one, in order to preclude negative interest rates. The argument was that investors could choose to keep their money under a proverbial mattress rather than on deposit earning a negative interest rate. But the spate of negative interest rates in recent history suggests there aren't enough mattresses under which to keep funds safe from negative interest rates. In more economic terms, banks clearly provide services for which depositors are willing to pay in the form of below-market interest rates, even if that means accepting negative nominal interest rates.

When choosing a functional form for $\phi(\tau)$, two conditions seem natural.

Criterion 1: We'd like a discount function that approaches zero in the limit as time increases. That is, $\lim_{\tau \rightarrow \infty} \phi(\tau) = 0$, where ϕ is the discount function and τ is the time until the relevant cash flow is received.

Criterion 2: To prevent arbitrage, we also require $\lim_{\tau \rightarrow 0} \phi(\tau) = 1$. Otherwise, an investor would be able to purchase (or sell) a zero-coupon bond an instant before maturity and earn a riskless profit well in excess of the riskless rate.

In our experience, exponential functions are useful for this purpose, and they clearly satisfy criteria 1 (C1) and 2 (C2).

For example, if we specify yields in terms of continuously compounded rates, we could use the general specification

$$\phi(\tau) = e^{-h(\tau)}$$

where $h(\tau)$ is a continuous function of τ .

As long as $h(\tau)$ is bounded, $\phi(\tau)$ will satisfy C1 and C2 above. Actually, boundedness is too strong a condition in this case. For example, if $h(\tau)$ is a polynomial in τ , it will also satisfy C1 as all polynomials increase at a slower rate than the exponential function. Nevertheless, the concept of an unbounded yield curve presents difficulties with economic interpretation if not necessarily with the mathematics. It's also nice, if not necessary, for $h(\tau)$ to have a limit as $\tau \rightarrow \infty$. (Note that polynomials do not satisfy this criterion.)

Depending on the functional form used for $h(\tau)$, there can be a number of advantages in specifying a functional form for the yield curve rather than for the discount factor function directly. In addition to the ease with which conditions C1 and C2 can be satisfied, specifying a functional form for the zero-coupon yield curve tends to offer greater intuition to the analyst during the curve-fitting exercise. For example, if a particular attempt to fit the curve

is producing modeled yields that are systemically too high for bonds maturing between seven and 10 years, it's typically easier to ascertain the desired changes to make when specifying a yield curve than it is when specifying a discount factor function that may contain upward of 10 constituent functions, all specified in terms of zero-coupon bond prices.

Ultimately, however, the choice is a function of the quality of the fit and of the ease with which the analyst is able to implement the model.

WEIGHTS

Thus far, we've maintained the assumption that the error terms have identical distributions, with zero means and with variances equal to one another. If there are reasons to believe that the variances of the error terms are not equal, we can modify our approach to accommodate WLS. In other words, we can choose parameters to minimize:

$$\sum_{i=1}^N \frac{\varepsilon_i^2}{U_i}$$

where U_i is the weight assumed to be proportional to the variance of ε_i .

In our experience, there are a few criteria that could help guide the choice of weights in this context:

- **Basis point values:** Some analysts take the view that bonds with large basis point values have prices that are measured with less precision and therefore with greater error than bonds with smaller basis point values. In particular, if the variances of the error terms for the fitted yields were assumed to be equal, then the variances of the fitted prices would be roughly proportional to the basis point values of the bonds.
- **Deliverability:** As discussed in Chapter 7 on future delivery options, bonds that are deliverable into a futures contract may experience greater price volatility than similar bonds that are not deliverable.
- **Repo specialness:** Bonds that are experiencing episodic specialness in the repo market have a tendency to have greater spot price volatility than other bonds.

Ultimately, the heteroscedasticity of the error terms is an empirical matter. We suggest analysts use judgment in deciding what forms of heteroscedasticity are likely to be present in the markets they're analyzing.

SETTING UP THE OPTIMIZATION

Reflecting the ideas outlined above in a numerical optimization, the key is to minimize an error term by changing the parameters controlling the discount factor curve *at the same time and in the same optimization* as the regression coefficients versus external explanatory variables, such as issuance size. Hence, the general approach is to minimize an expression like the following by changing the parameters determining the discount factor curve df and the regression coefficients $\gamma(k)$:

$$\sum_{b \in \text{Bonds}} \left| \sum_{i \in CF(b)} CF(i) \times df(t_i) + \sum_{k \in \text{External}} \gamma(k) \times k(b) - \text{Dirtyprice}(b) \right|^{1 \text{ or } 2 \text{ or } \dots}$$

where:

- $CF(i)$ is the i -th cash flow of the bond b at time t_i
- df is the discount factor curve
- $\gamma(k)$ is the regression coefficient versus external variable k (such as issuance size).

In case of exponential splines being used as functional form for the discount factor curve, i.e. $df(t) = \sum_{j=1}^n \beta_j \times e^{-jt\alpha}$, the parameters β_j and α are subject to the optimization – *at the same time* as the $\gamma(k)$.

Hence, the *output* of the optimization will be a fitted curve df (in the case of exponential splines given by the parameters β_j and α) and the regression coefficients $\gamma(k)$ versus external driving forces.

When setting up the optimization, the analyst faces a number of important decisions regarding the *input*:

- the choice of the sets “Bonds” and “External”
- the choice of the functional form for df , and in case of exponential splines of their number n
- the choice of the final exponent (of the absolute spread between fitted and actual dirty price). While quadratic error terms seem to be considered as standard, in our experience, this often results in outliers gaining too much influence on the result of the optimization; we therefore recommend trying the exponent 1 instead of 2 as well.

These choices fall to the discretion of the analyst and require careful adjustment to the markets analyzed and analytic goals.

Example: Fitting the German Bund Curve

The Setting: German Bunds, Using Size, Age, and Deliverability as Additional Conditioning Variables

The best way to illustrate these concepts is to work through a specific example. In this case, we use the closing prices of coupon-paying German Bunds, OBLs, and Schätze on 7 Dec 2012. As additional independent variables, we'll include:

- **Time since issuance:** New issues may be in greater demand than older issues.
- **Issue size:** To achieve benchmark status, a bond needs to be issued in sufficient size.
- **Deliverability:** Bonds that are deliverable into futures contracts, in particular the cheapest-to-deliver (CTD), may be priced differently from other bonds.

Note that theoretical arguments can be made for the coefficients on these three variables to be positive or negative. For example, benchmark issues may be easier to sell short, making them less expensive than otherwise comparable issues. Likewise, if a CTD issue is easier to sell short, its price might be less than it would be otherwise. So ultimately, the signs of the coefficients will be determined by the data.

Our list of explanatory variables isn't meant to be exhaustive. There are reasons to include repo specialness as a variable. It's also useful to consider whether recent CTDs might maintain residual richness or cheapness for a period and whether likely future CTDs might contain some anticipatory richness or cheapness. For illustrative purposes, however, we'll focus on the explanatory variables listed above.

Figure 8.1 shows the term structure of yields for coupon-paying German Bunds as of the close of trading on Friday, 7 December 2012. We exclude bonds with times to maturity of less than one year, as we find money markets are best modeled separately.¹

¹Money market instruments, including bonds with less than a year to maturity, require a greater degree of analytical precision than longer-dated bonds, for a number of reasons. For example, money market instruments are particularly sensitive to central bank policy, and at times we observe sharp discontinuities between instruments maturing before particular central bank meeting dates and those maturing after.

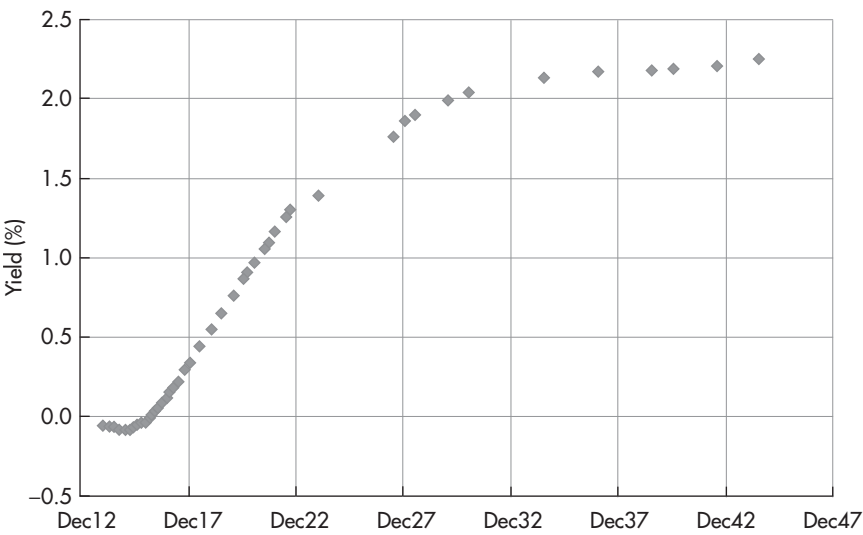


FIGURE 8.1 Bund yields as of 7 Dec 2012.
Source: Data – Bloomberg; Chart – Authors.

TABLE 8.1 Deliverable Issues into March 2013 Futures Contracts

Schatz	Bobl	Bond	Buxl
BKO 0% 12-Dec-14	DBR 4% 4-Jan-18	DBR 2% 4-Jan-22	DBR 4.75% 4-Jul-40
DBR 3.75% 4-Jan-15	OBL 0.5% 13-Oct-17	DBR 1.75% 4-Jul-22	DBR 4.25% 4-Jul-39
OBL 2.5% 27-Feb-15	DBR 4.25% 4-Jul-18	DBR 1.5% 4-Sep-22	DBR 3.25% 4-Jul-42
OBL 2.25% 10-Feb-15			DBR 2.5% 4-Jul-44

Table 8.1 shows the issues that are deliverable into the Schatz, Bobl, Bund, and Buxl futures contracts expiring in March 2013.

We define our deliverability variable in such a way that the deliverability variable is not simply one if a bond is deliverable and zero otherwise. Rather, this variable is set equal to the price value of a basis point for the bond if the bond is deliverable, and it's set to zero otherwise. As a result, the coefficients for these variables have the interpretation of being the number of basis points by which an issue tends to be rich when it is the deliverable issue into one of the futures contracts.

Choosing a Functional Form for the Yield Curve

The next step in implementing this approach is specifying a suitable functional form for $h(\tau)$.

This choice depends on both the market and the goals of the analysis, and it is advisable to try and compare the results of several different functional forms.

- Exponential splines described above offer the advantage of being naturally linked to discount factor curves and of having a limit for $\tau \rightarrow \infty$. Hence, if *extrapolations* are part of the goals, exponential splines could well be the only suitable alternative. For example, if the task is to price the first 50Y bond of a government, which has only issued maturities up to 30Y so far, polynomials are no alternative. Having decided to use exponential splines leads to the next decision on how many to use, with the answer depending again on the market and analysis goals. For typical bond markets, seven could be a good starting point, with too few splines causing a too sticky curve and too many splines following each outlier too closely.
- Cubic splines consist of a series of cubic polynomials between carefully chosen knot points (e.g. 0Y, 1Y, 2Y, etc.), constructed so as to ensure continuity in the function, first and second derivative. They are more suitable to reflect changes in the first derivative of the yield curve. For example, if the market assumptions about monetary policy result in interest rates rising between 0Y and 1Y, falling between 1Y and 2Y before rising again, cubic splines are in a better position to capture this behavior than exponential splines. Having decided to use cubic splines leads to the next decision on how many and which knot points to use. Since changes in the first derivative of the yield curve are more common at the front end influenced by monetary policy expectations, it is typically advisable to increase the distance between knot points with maturity.
- If cubic splines provide unsatisfactory results, the analyst can specify different functional forms between the knot points and obtain maximum smoothness, defined here as the integral of the square of the second derivative of the function.

For this example, we choose cubic splines and place the knots between each cubic spline at maturities (in years) of: 0, 0.5, 1.0, 1.5, 2, 5, 7, 10, 12, 15, 20, 30, and 35.

The details of this calculation are explained in the box and the resulting regression residuals are shown in Figure 8.2.

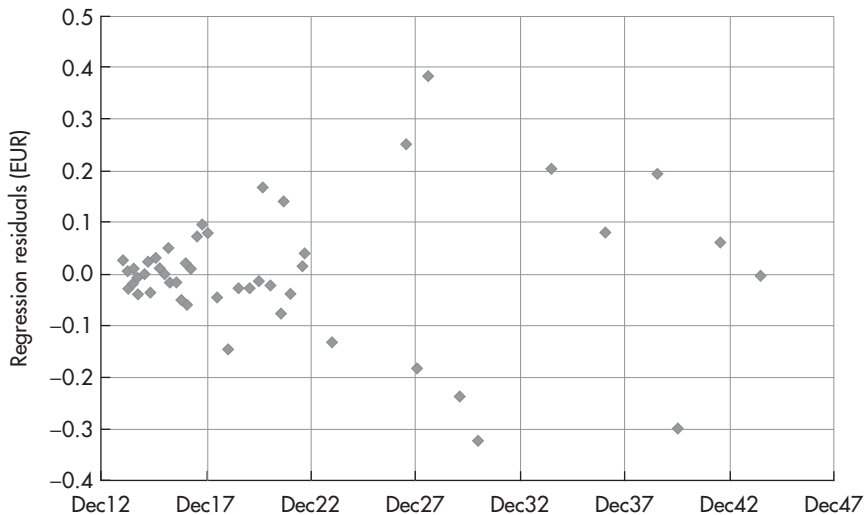


FIGURE 8.2 Bund regression residuals expressed in EUR.

Source: Authors.

Technical Points for Calculating Cubic Splines

While determining the parameters of each cubic polynomial is a straightforward linear algebra problem (solve for the parameters to fulfill the continuity in the function, first and second derivatives), this calculation needs to be repeated in every step of the optimization outlined above. Hence, it is important to use a numerically quick set-up, e.g. the following routine:

Given the knot points (x_i, y_i) define h_i as the distance $x_i - x_{i-1}$ between consecutive maturities and the tridiagonal $n \times n$ -matrix A by placing 2 in the main diagonal, $\frac{h_i}{h_i + h_{i+1}}$ to the left and $1 - \frac{h_i}{h_i + h_{i+1}}$ to the right (in the first and last row, all entries are 0 except for 2 in the main diagonal).

Note that the matrix A is independent of the values y_i , which change during the optimization. Hence, the inverted matrix A^{-1} only needs to be calculated once and can be used for all steps of the optimization, improving the speed significantly.

With $d_i := \left(\frac{y_{i+1} - y_i}{h_{i+1}} - \frac{y_i - y_{i-1}}{h_i} \right) \frac{6}{h_i + h_{i+1}}$ (and $d_1 := d_n := 0$) and $M := A^{-1}d$, the spline value at time to maturity $t \in [x_i, x_{i+1}]$ is given by $\frac{M_i(x_{i+1} - t)^3}{6h_{i+1}} + \frac{M_{i+1}(t - x_i)^3}{6h_{i+1}} + \left(y_{i+1} - \frac{M_{i+1}h_{i+1}^2}{6} \right) \frac{t - x_i}{h_{i+1}} + \left(y_i - \frac{M_ih_{i+1}^2}{6} \right) \frac{x_{i+1} - t}{h_{i+1}}$

Assessing the Quality of the Fit

The average absolute value of the residuals is 9 cents. When interpreting this figure, it's useful to note that there does appear to be some evidence of heteroscedasticity in the residual terms. In particular, the magnitudes of the residual terms appear to be relatively large for longer maturities. With this in mind, we plot the residuals in terms of basis point values for the bonds in Figure 8.3.

In this case, the average absolute value of the residuals is 0.9 bp, with only one residual term less than -2 bp and only two residual terms greater than 2 bp. The largest positive residual has a magnitude of only 2.3 bp, and the largest negative residual has a magnitude of only 2.2 bp. With these results in mind, the quality of the fit appears to be satisfactory.²

Interpreting the Regression Coefficients

The coefficients for the size, age, and deliverability variables are shown in Table 8.2.

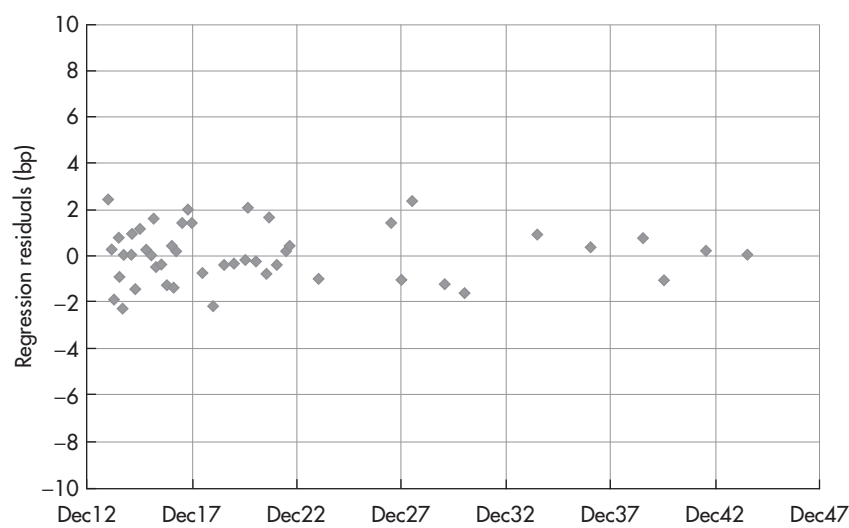


FIGURE 8.3 Bund regression residuals expressed in basis points.
Source: Authors.

²Given that the residuals appear somewhat heteroscedastic when expressed in terms of price and less so when expressed in terms of basis points suggests further analysis may be warranted to assess the merits of WLS in this case.

TABLE 8.2 Bund Regression Coefficients

Independent variable	Coefficient estimate
Amount outstanding (EUR bn)	EUR 0.025
Issue age (years)	0.027 years
Schatz deliverable? (bp)	3.6 bp
Bobl deliverable? (bp)	2.8 bp
Bund deliverable? (bp)	2.1 bp
Buxl deliverable? (bp)	2.4 bp

The coefficient for the amount outstanding suggests that larger issues tend to be more expensive, which would be consistent with the existence of a benchmark effect in this market. In particular, an additional EUR 1 bn outstanding corresponds to an additional 2.5 cents in the value of the bond.

The coefficient for the age of the bond suggests that bonds tend to cheapen with age. This also is consistent with a benchmark effect, and it's perhaps consistent with a repo effect as well. In particular, in the German market, bonds tend to cheapen at a rate of 2.7 cents per year, all else being equal.

The results are broadly similar for the deliverability variables. In particular, issues that are deliverable into the Schatz, Bobl, Bund, and Buxl contracts tend to have yields that are lower by 2 to 4 bp than other bonds. The fact that the premium for the Schatz contract is the largest perhaps shouldn't be surprising, since the basis point values are so small for the bonds that are deliverable in the Schatz contract.

Statistical Analysis of Rich/Cheap Figures

In our experience, the rich/cheap indicators for individual bonds generally tend to revert around a mean of zero, though this isn't always the case. For example, in some cases, bonds are structurally rich, and the rich/cheap indicator will revert around a number that reflects this structural richness.

In other cases, the rich/cheap indicator won't exhibit any particular mean reversion at all. While we would be reluctant to assume that the rich/cheap indicator is non-stationary, given the economic implications of this inference, the failure for a time series to exhibit clear mean reversion makes it difficult for us to estimate the speed of mean reversion, even if we have persuasive reasons *a priori* for assuming the indicator to revert around a mean.

Z-Scores and T-Stats

In our experience, many analysts model the mean reversion of rich/cheap indicators by calculating the number of standard deviations by which the